BRENDA
The Comprehensive Enzyme Information System

Information on EC 3.2.1.8 - endo-1,4-beta-xylanase

TEMPERATURE RANGE	ORGANISM	UNIPROT	COMMENTARY	LITERATURE
10 - 55	Pseudopedobacter saltans	F0S6Y3	recombinant enzyme, activity range, profile overview	752084
10 - 65	Sphingobacterium sp. TN19	D8L2Y2	the enzyme shows over 65% of its maximum activity at 20-28°C and about 40% at 10°C	707250
10 - 80	Pediococcus acidilactici	-	native extracellular enzyme, over 65% activity within this range, profile overview	750814
10 - 90	Demequina sp. JK4	B9VSZ3	enzyme assay range, 50% of maximal activity at 45°C and 60°C	709491
100	Thermomyces lanuginosus	-	40% of maximum activity	742766
20 - 50	Penicillium griseofulvum	-	-	707567
20 - 60	Cladosporium sp.	A0A346PXS9	20°C: about 40% of maximal activity, 60°C: about 35% of maximal activity	750679
20 - 70 ⊟	Aspergillus sulphureus	Q3S401	50% of maximal activity at 20°C, 20% at 70°C	707197
	Nesterenkonia xinjiangensis	D1KJJ7	activity range, profile, overview	707216
20 - 80 ⊟	Bispora sp. MEY-1	D6MYS9	-	707258
	Paenibacillus sp.	B6VF01	activity range	717404

	Streptomyces avermitilis	Q9X584	purified heavier protein band (42.8 kDa), more than 50% of maximum activity	698373
20 - 90 ⊟	Thermoanaerobacterium saccharolyticum	P36917	the enzyme shows 84.1%, 91.5% and 97.3% relative activities at 40, 50, and 60 °C, respectively, compared to the maximum activity at 70°C. At lower temperatures, such as 20 and 30°C, the enzyme exhibits more than 40% relative activity. In addition, the enzyme is active at high temperatures of 80 and 90°C, with relative activities of 52.8 and 30.2%, respectively	775899
	Trichoderma reesei	-	activity range, profile, overview	707911
25 - 60	Dickeya chrysanthemi	-	-	655834
25 - 70	Acidomyces acidophilus	-	-	707765
25 - 75	Bacillus licheniformis	-	activity range, profile overview	752334
30 - 45 ⊟	Duganella sp. PAMC 27433	-	the enzyme shows more than 50% activity between 30 and 45°C. The enzyme displays 12% of its maximum activity at 4°C and 20% activity at 8°C	769515
	uncultured bacterium	-	approximately 80% activity is maintained in intervals of 30 and 45°C	775836
30 - 50 ⊟	Massilia sp. RBM26	A0A0S2I9J6	more than 60% of maximum activity	738866

	Penicillium occitanis (nom. inval.)	-	more than 75% of the activity at 30°C, rapidly decrease in activity above 50°C	718338
	Thermochaetoides thermophila	Q6UN40	-	701091
	uncultured bacterium	A0A3G9FT71	30°C: about 60% of maximal activity, 50°C: about 50% of maximal activity	750245
30 - 55 ⊟	Bacillus subtilis	P18429	activity range, inactive at 60°C, temperature profile of recombinant His-tagged wild-type and mutant enzymes, overview	710466
	Kitasatospora sp.	A0A3T0YV67, A0A3T0YV60	50% of maximal activity within this range, profile overview	750084
30 - 60	uncultured bacterium	A0A127JBT0	more than 60% of maximum activity	744542
30 - 65	Kitasatospora sp.	A0A3T0YV67, A0A3T0YV60	50% of maximal activity within this range, profile overview	750084
30 - 70 ⊟	Aspergillus fumigatus	-	-	682991
	Aspergillus niger	-	activity range, profile, overview	710686
	Bacillus subtilis	A0A172MAU1	30°C: immobilized enzyme shows about 50% of maximal activity, soluble enzyme shows about 40% of maximal activity. 70°C: immobilized enzyme shows about 65% of maximal activity, soluble enzyme shows about 50% of maximal activity	750593

	Cellulomonas flavigena	A1XM14	-	697503
	Kocuria sp. Mn22	B8XY24	activity range	709503
	Paenibacillus sp. E18	-	profile, overview	707230
	Pleurotus ostreatus	A0A2U9Q8K4	30°C: about 50% of maximal activity, 70°C: about 35% of maximal activity	750334
	Streptomyces thermocyaneoviolaceus	Q9RMM4	native enzyme, activity range, profile, overview	707809
	Trichoderma reesei	B2CNY5	30% of maximal activity at 70°C	709576
	Trichoderma sp.	-	activity range, isozyme EX2	707762
30 - 75 ⊟	Aspergillus fumigatus	-	more than 60% of maximum activity is retained at temperatures ranging from 30 to 75Â°C	769929
	Streptomyces sp.	-	about 50% of maximal activity at 50°C and 75°C	654144
	Trametes hirsuta	-	30°C: about 50% of maximal activity, 75°C: about 30% of maximal activity	171629
30 - 80	Streptomyces halstedii	-	30°C: about 30% of maximal activity of enzyme form Xys1L, about 50% of maximal activity of enzyme form Xys1S, 80°C: about 15% of maximal activity of enzyme form Xys1L, about 35% of maximal activity of enzyme form Xys1S	171639
30 - 85	Acetivibrio thermocellus	-	over 50% of maximal activity within this range, profile overview	738886

30 - 90 ⊟	Thermochaetoides thermophila	G0S9G3	activity range, profile overview	750799
	Thermotoga maritima	-	activity range of mutant hybrid enzyme, profile, overview	710126
	Trichoderma reesei	-	activity range of wild-type and mutant enzymes, profile, overview, 30% of maximal activity at 75°C	710126
		-	recombinant enzyme, activity range, 45% of maximal activity at 80°C, profile, overview	707415
	Trichoderma sp.	-	activity range, isozyme EX1	707762
35 - 100	Bispora sp. MEY-1	D0QF43	activity range, 80% of maximal activity at 75-90°C, 20% at 100°C, and 10% at 40°C, profile, overview	708455
35 - 60	Cellulomonas flavigena	-	more than 80% of maximum activity	746350
35 - 68	Aspergillus niger	-	more than 68% of maximum activity	731024
35 - 75 ⊟	Streptomyces thermocarboxydus	C6ZHB0	activity range	709555
	uncultured microorganism	E2DQY5	activity range	717134
37 - 60	Acetivibrio thermocellus	-	more than 50% activity between 37 and 60°C	750432
40 - 100	[Caldibacillus] cellulovorans	Q9L8L8	recombinant N-terminal + catalytic domains	656718
40 - 50 ⊟	Bacillus licheniformis	B5SYI8	-	697502

	Bacillus subtilis	B9ZZN9	activity sharply decreases below 30°C and above 60°C	699055
40 - 55	Aureobasidium melanogenum	-	more than 75% activity between 40 and 55°C	776226
40 - 60 ⊟	Bacillus subtilis	Q45VU2	transxylosylation and hydrolysis activities are optimal at temperatures of 60°C and 40°C, respectively	696821
	Fusarium oxysporum	-	more than 80% of maximal activity at 40°C and at 60°C	171637
	Penicillium citrinum	Q2PGY1	recombinant enzyme, over 50% of maximal activity within this range, profile overview	752115
	Trichoderma harzianum	-	40°C, about 70% of maximal activity, 60°C: about 30% of maximal activity, 20000 Da xylanase	171613
	Trichoderma sp. TP3-36	-	more than 50% activity between 40 and 60°C	776215
40 - 63	Aspergillus flavus	B8NJ86	over 50% of maximal activity within this range, profile overview	750988
40 - 65	Alicyclobacillus sp.	D3VWB5	XynA4 shows over 50% of maximal activity within this range	717947
40 - 70 ⊟	Acetivibrio thermocellus	A3DJS9	more than 95% of maximum activity	737360
	Evansstolkia leycettana	-	the enzyme shows about 50% activity at 40 C and 40% activity at 70 C. At room temperature (23 C), the enzyme is almost	774449

			inactive (approximately 1% activity)	
	Fusarium proliferatum	-	89% of maximal activity at 60°C and 14% at 70°C	657201
	Streptomyces lividans	-	40°C: about 50% of maximal activity, 70°C: about 25% of maximal activity	171608
	Streptomyces sp. T7	-	the enzyme shows approximately 60% of the maximum activity when assayed between 40 and 70°C, while more than 40% of the enzyme activity below 20°C or above 80°C	774465
40 - 75	Bacillus sp. (in: firmicutes)	-	40°C: about 25% of maximal activity of enzyme from strain W1, about 30% of maximal activity of enzyme from strain W3, 75°C: about 30% of maximal activity, enzyme from strain W1 and W3	171621
40 - 80 ⊟	Bacillus sp. (in: firmicutes)	-	40°C: about 20% of maximal activity, 80°C: about 45% of maximal activity	171656
	Caldicellulosiruptor saccharolyticus	-	40°C: about 30% of maximal activity, 80°C: about 20% of maximal activity	171655
	Niallia circulans	-	-	678008
	Paenibacillus campinasensis	Q2I6W5	activity range, correlation with pH, overview	718337
	Saccharopolyspora pathumthaniensis	E1APH5	activity range, profile, overview	717401

	Thermochaetoides thermophila	G0S9G3	40°C: about 40% of maximal activity, 80°C: about 50% of maximal activity, substrate: xylan	750799
		G0S9G3	40°C: about 50% of maximal activity, 80°C: about 50% of maximal activity, substrate: carboxymethyl cellulose	750799
40 - 85 ⊟	Acidothermus cellulolyticus	A0LR95	activity range, inactive below and above, recombinant enzyme, profile overview	750497
	[Caldibacillus] cellulovorans	Q9L8L8	recombinant catalytic domain	656718
40 - 90 ⊟	Bacillus amyloliquefaciens	-	40°C: about 50% of maximal activity, 90°C: about 85% of maximal activity	171646
	Bacillus sp. (in: firmicutes)	-	activity range	717105
	Rhizomucor miehei	-	activity range	717022
	Talaromyces purpureogenus	A0A2H4RAC5	recombinant enzyme, over 80% of maximal activity within this range	749635
45 - 55	Fusarium oxysporum	-	more than 70% of maximum activity within this range	663671
45 - 75	Bacillus pumilus	-	at least 50% activity between 45 and 75°C	769914
45 - 85	Geobacillus stearothermophilus	-	45°C: 28% of maximal activity, 85°C: 74% of maximal activity	171651
5 - 35	Pseudoalteromonas haloplanktis	Q8RJN8	60% of maximal activity at 5°C	656075
50 ⊟	Streptomyces coelicolor	Q8CJQ1	98% of maximum activity	737481

	Thermoascus aurantiacus	-	assay at, about 70% of maximum activity	742959
	Thermoascus aurantiacus var. levisporus	-	80% of maximum activity	743330
50 - 100 ⊟	Bacillus subtilis	-	50°C: about 50% of maximal activity, 100°C: maximal activity	751229
	Thermobifida alba	-	50°C: about 40% of maximal activity, 100°C: about 65% of maximal activity	171636
50 - 55	Alternaria mali	-	-	695683
50 - 65	Acrophialophora nainiana	-	rapid decline of activity at 65°C	655620
50 - 70 ⊟	Aspergillus niger	-	activity at 50°C is about 20% higher compared to activity at 70°C	737497
	Rhodothermaceae bacterium RA	A0A1V0DJ74	the enzyme retains more than 60% of its maximal activity at a temperature ranging from 50 to 70Â°C	775585
	Streptomyces lividans	-	50°C: about 60% of maximal activity, 70°C: about 75% of maximal activity	171606
	Thermothelomyces heterothallicus	-	activity at 50°C is about 10% compared to the activity at 70°C	737497
		-	activity at 50°C is about 25% compared to the activity at 70°C	737497
	Thermothelomyces thermophilus	-	activity at 50°C is about 5% compared to the activity at 70°C	737497
	Trichoderma reesei	-	activity at 50°C is slightly higher than activity at 70°C	737497

50 - 75	Bacillus sp. (in: firmicutes)	-	50°C: about 50% of maximal activity, 75°C: about 60% of maximal activity	171621
50 - 80 ⊟	Acidothermus cellulolyticus	A0LR95	about 60% activity at 50°C, about 750% activity at 60°C, 100% activity at 70°C, about 90% activity at 80°C, no activity at 90°C	750497
	Bacillus sp. (in: firmicutes)	-	50°C: about 30% of maximal activity, 80°C: about 20% of maximal activity, enzyme form strain W4	171621
50 - 85 ⊟	Thermomyces lanuginosus	-	about 50% of maximal activity at 50°C and at 85°C	171643
	Thermopolyspora flexuosa	-	recombinant His-tagged enzyme, inactive at 90°C	707229
55 - 60	Pseudozyma hubeiensis	-	PhX20	707766
60 - 100 ⊟	Acidothermus cellulolyticus	-	activity range, pH-dependent: at 100°C activity occurs at pH 4.0-6.0, at 60°C activity occurs at pH 5.0-8.0, overview	717123
	Saccharolobus solfataricus	-	-	679721
60 - 65	Pseudozyma hubeiensis	-	PhX33	707766
60 - 80 ⊟	Bacillus sp. (in: firmicutes)	-	60°C: about 50% of maximal activity, 80°C: about 80% of maximal activity	171645
	Thermobifida fusca	Q47QL8	more than 60% of maximum activity, recombinant protein, both with and without His-tag	737932

60 - 90	Thermoclostridium stercorarium	-	60°C: about 50% of maximal activity, 90°C: about 20% of maximal activity	171661
65 - 75	Penicillium canescens	C3VEV9	more than 80% of maximum activity	729182
65 - 85	Streptomyces ipomoeae	MK331807	65°C: about 45% of maximal activity, 85°C: about 50% of maximal activity	750086
70 ⊟	bacterium enrichment culture clone XylI8B8	H9ZGD1	65% residual activity	730872
	Geobacillus stearothermophilus	-	18% of maximum activity	738190
70 - 100	Thermotoga petrophila	A5IL00	70°C: about 40% of maximal activity, 100°C: about 70% of maximal activity	748630
70 - 75	Thermomyces lanuginosus	-	-	171643
70 - 85	Thermoascus aurantiacus	-	70°C: about 70% of maximal activity, 85°C: about 10% of maximal activity	171601
75	Thermomyces lanuginosus	-	assay at	742214
80 ⊟	Geobacillus stearothermophilus	-	9% of maximum activity	738190
	Halalkalibacterium halodurans	-	80% of maximum activity	731022
	Thermomyces lanuginosus	-	rapid loss of activity above	742165
80 - 90	Laetiporus sulphureus	-	activity decreases rapidly at temperatures over 80°C, but still retains approximately 70% relative activity at 90°C	699053
80 - 95	Saccharolobus solfataricus	Q97YG7	80°C: about 40% of maximal activity,	692258

			95°C: about 40% of maximal activity	
85	Thermomyces lanuginosus	-	more than 50% of maximum activity	646829
additional information ⊟	Carica papaya	Q8GTJ2	rapid decline in activity above 45°C	655665
	Trichoderma reesei	-	temperature-dependence of activity of wild-type and mutant enzymes	656334